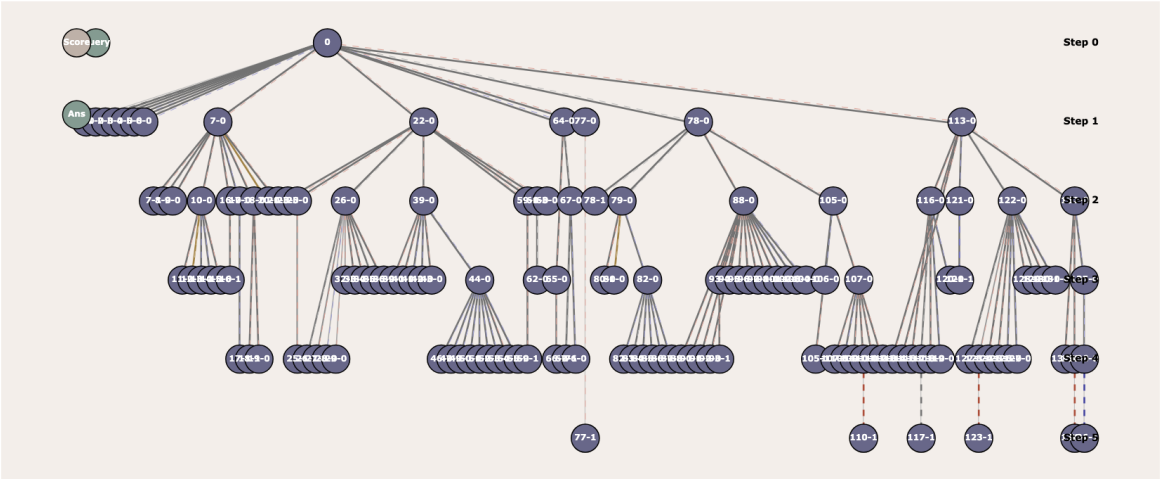

Figure 12: Visualization results of tree structure of a response from **DeepSeek-32B** on **GPQA** dataset extracted using LCoT2Tree. The reasoning trees contain many high out-degree nodes, indicating that the model often revisits and elaborates on complex concepts.

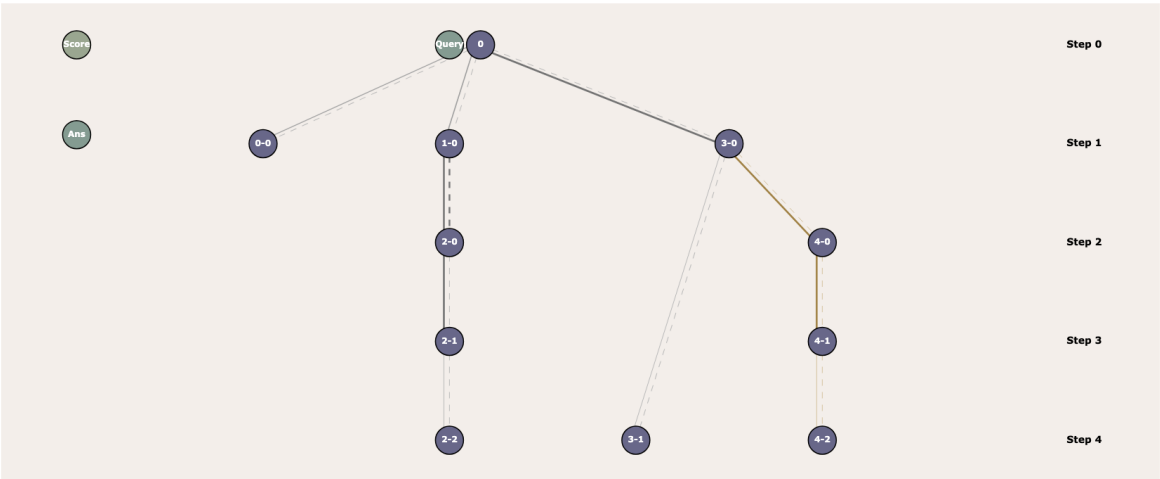

Figure 13: Visualization results of tree structure of a response from **DeepSeek-32B** on **MMLU-Pro** dataset extracted using LCoT2Tree. The reasoning trees contain fewer nodes and minimal branching, indicating a more direct and deductive reasoning style with less exploration.

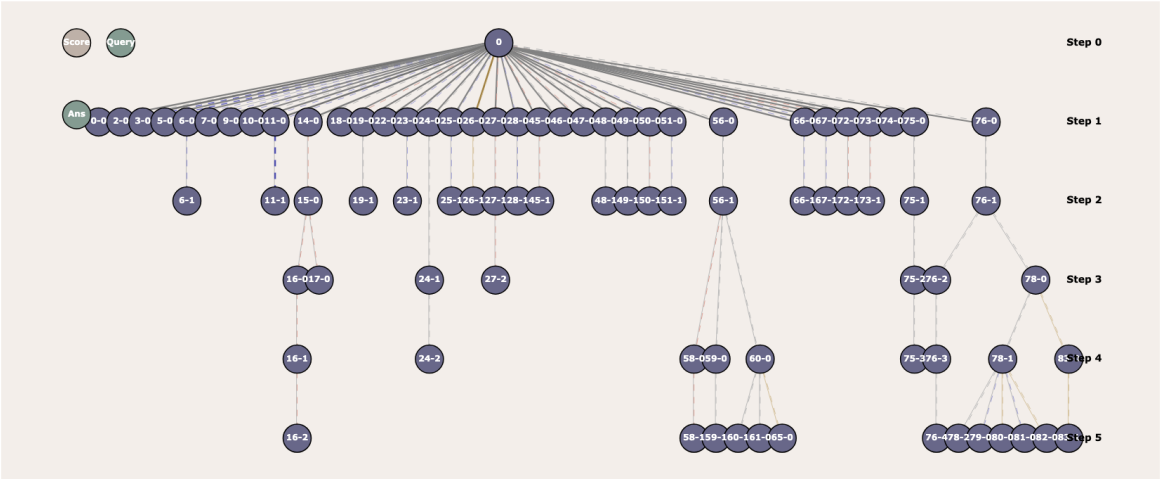

Figure 14: Visualization results of tree structure of a response from **DeepSeek-R1** on **MATH** dataset extracted using LCoT2Tree. It exhibits similar behavior to DS-32, but with an important distinction: it tends to truncate detailed exploration earlier and backtrack to beginning steps more quickly to optimize its reasoning path.

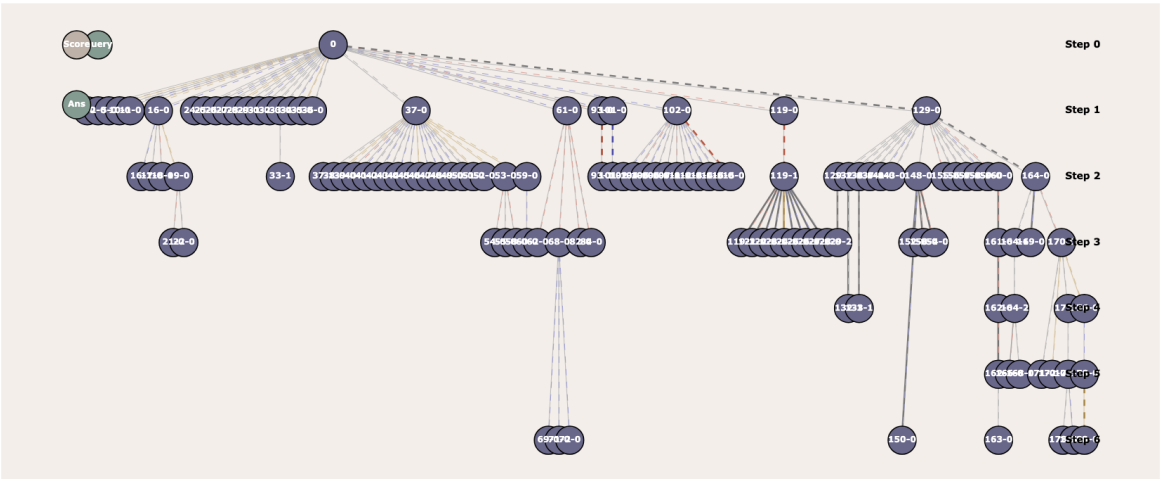

Figure 15: Visualization results of tree structure of a response from **QwQ-32B** on **MATH** dataset extracted using LCoT2Tree. QwQ-32B mirrors the behavior of DeepSeek-32B to some extent, but differs in how it allocates attention in the latter stages of reasoning.

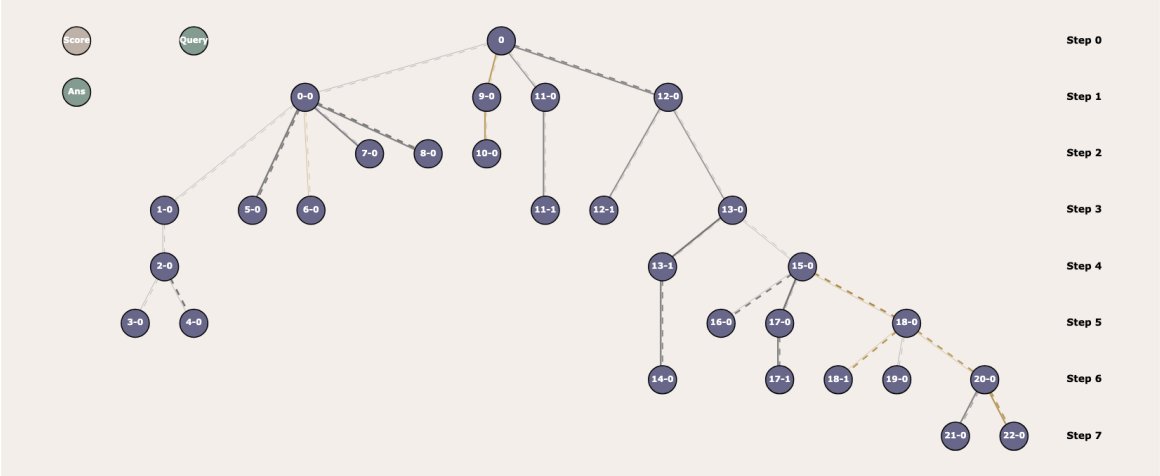

Figure 16: Visualization results of tree structure of a response from **Seed-1.5-Thinking-pro** on **MATH** dataset extracted using LCoT2Tree. The reasoning trees contain fewer nodes and branches, forming simpler structures.

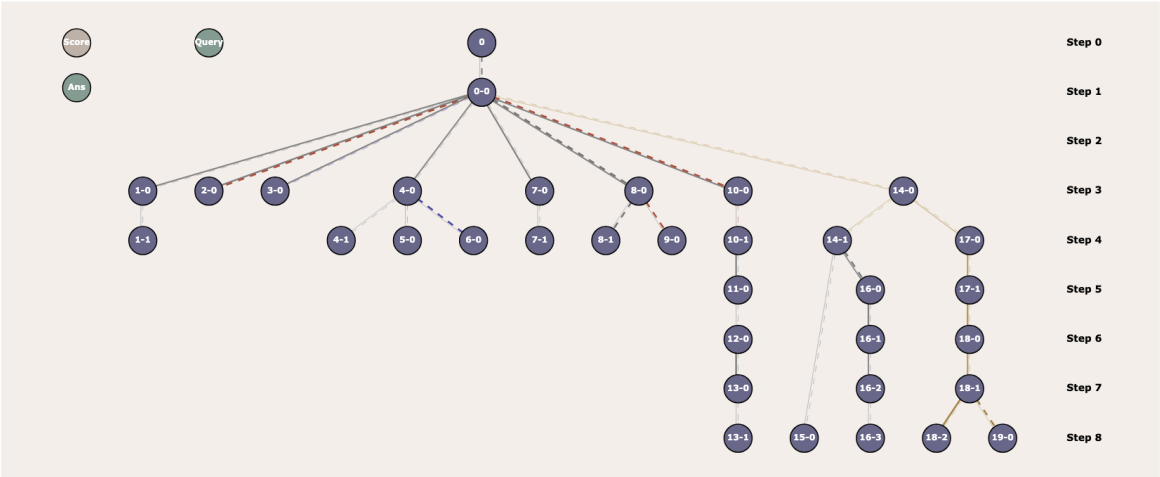

Figure 17: Visualization results of tree structure of a response from **Grok-3-mini-beta** on **MATH** dataset extracted using LCoT2Tree. The reasoning trees contain fewer nodes and branches, forming simpler structures.